Section 7: Failure Modes and Effects Analysis

FMEA Scope

This dFMEA covers each step for the user of the Scott Russell linkage plunger device, from opening the sterile package to pushing the driving lever arm to deliver the dose. Using a functional block diagram, we identified 13 failure modes by determining input and output failures of each subcomponent. Each was scored for severity, occurrence, and detectability, and ranked by RPN. The highest risks involve direct harm to the patient's eye through methods such as: excess force applied by the user, improper sterilization, and inconsistent injection rates. Mechanical failures, such as a jammed lever or slider, were lower risk because they would likely be noticed before reaching the patient. Long term wear was also deemed high risk due to the low likelihood of detection and its ability to magnify any of the aforementioned issues. Recommended mitigations will guide our next prototype iteration, and after each design change, we will re-score affected failure modes, determine whether we reduced RPNs through testing, and add new failure modes if needed.

FMEA Table

Using a functional block diagram as a guide, evaluate your solution and identify at least ten different potential failure modes. For each failure mode, you will need to explore what happens if there is a loss in the input or output of a subcomponent. As the design evolves, this table needs to be updated to address changes to subcomponents of the design.

Process/ Step/Input /Output	Potential Failure Mode	Potential Failure Effect	S	Potential Causes	O	Current Controls	D	RPN	Recommended Mitigation
Insert the commercial 1 mL syringe into the device	The opening of the sterilizable housing is not compatible with the syringe	Device cannot be used on the syringe	6	Device was not tested with that brand/size of syringe	2	Design device to be compatible across all major 1 mL syringe brands	1	12	Test fit with common 1 mL syringe brands used for intravitreal injections
Insert the syringe into the opening of the sterilizable housing	Syringe is not securely locked during the process and falls out	Can harm the patient or healthcare provider	5	Syringe quality is poor, human factors error, poor machining tolerance of device	4	Manufacture device with high tolerances to ensure good fits and use high quality syringes	2	40	Perform a pull test before injection to verify secure fit
Push the driving arm lever forward until it cannot be	The driving arm lever does not move	Syringe can no longer be used; potential wastage of	4	The screw connecting the driving lever arm and Scott Russell	2	Design screw to be easily accessible for replacement	2	16	Use corrosion-resistant material (stainless steel) for the screw

pushed any further		drug and device		coupler link is jammed or rusted					and check lever mobility before injection
Push the driving arm lever forward until it cannot be pushed any further	The output slider (plunger-drive carriage) does not touch the plunger	Syringe can no longer be used - potential wastage of drug and device	4	Either plunger length is too short or inadequate force is applied by the user	2		2	16	Train users on proper operation of device**????**
Push the driving lever arm forward until it cannot be pushed any further	The output slider (plunger-drive carriage) does not slide forward	Syringe can no longer be used - potential wastage of drug and device	4	Lack of lubrication between slider and linear guide rail, high friction or jammed/rusty slider	2	Use materials that have low friction (PTFE on stainless steel)	2	16	Supply nontoxic lubricant for use with device
Push the driving lever arm forward until it cannot be pushed any further	The plunger does not push through with enough force	Delayed syringe administration	6	User does not apply enough force to push the lever	4	Confirm during testing the lever's mechanical advantage covers all potential high-viscosity formulations	6	144	Set the starting lever angle in the high mechanical advantage range
Push the driving lever arm forward until it cannot be pushed any further	User slides the linear guide rail instead of pushing the driving lever arm	Potentially inadequate pushing force leading to delayed administration	4	Human factors error, poor IFUs	2	Color code the location where thumb touches the lever arm to make easily distinguishable	4	32	Fix the rail to the housing so it can't be moved by hand
Push the driving lever arm forward until it cannot be pushed any further	User applies excess force	Can cause extreme harm to the patient including damage to tissue	10	Human factors error	6	Add stop that limits slider travel to the dose volume	6	360	Train user to push slowly
Push the driving lever arm forward until it cannot be pushed any further	The output slider (plunger-drive carriage) moves forward inconsistently	Delivery rate is inconsistent leading to poor absorption of drug by tissues;	8	Lack of lubrication, high friction	6	In the testing phase, test flow rate for consistency. Also use same low-friction materials and lubrication as #5.	6	288	Redesign with better tolerances to enable more consistent sliding

		ineffective treatment							
Remove device from sterilization pouch	Device did not get sterilized properly	Can introduce pathogens to patient's eye tissue	10	Due to cracks, crevices, or material incompatibilities with the sterilization protocol	3	Use materials validated for chosen sterilization method	8	240	Design without hidden crevices and with smooth surfaces
Hold and position the device for injection	Device makes it harder to do the procedure	Device increases procedure time	4	Device is clumsy to hold, ergonomic issues	8	Test with mock injections and mock procedures	2	64	Work with physicians / clinicians to reshape grip on device
Insert the tapered needle	Tapered needle falls off	Effects may range from no harm to infection in patient	6	Needle was not manufactured with proper plastic fitting	2	Use luer lock	2	24	Perform pull test before injection
Push the driving lever arm forward until it cannot be pushed any further	Output slider (plunger-drive carriage) or driving lever arm wears over time	Varying drug administration rate and effects vary from tissue damage to ineffective treatment	4	Natural wear due to the device being a mechanical device experiencing high forces	8	Use materials that don't face as much wear (stainless steel)	8	256	Set a maximum number of uses or sterilization cycles before replacement

S =severity; O=occurrence; D=Detectability; RPN=Risk Priority Number=S×O×D

Table that defines the Severity, Occurrence, and Detection Scale

Score:	Severity	Occurence	Detection
1-2	Does not fit seamlessly into clinical workflow (no impact)	Rarely occurs (less than 1 in 10000 uses)	Always noticed before use
3-4	Increases injection difficulty compared to not using the device	Occasionally occurs (about 1 in 1000 uses)	Easily noted during setup
5-6	Causes the procedure to fail.	Sometimes occurs (about 1 in 100 uses)	Noted only with careful inspection
7-8	Causes minor or reversible harm to patient (discomfort)	Frequently occurs (about 1 in 10 uses)	Nearly unnoticeable.
9-10	Can severely harm the patient	Almost always occurs (more than 1 in 2 uses)	Unnoticeable before risk occurs.